Linear mixed effects models

Data: flipped classrooms?

Data set has 375 rows (one per student), and the following variables:

- `professor`: which professor taught the class (1 -- 15)
- `style`: which teaching style the professor used (no flip, some flip, fully flipped)
- `score`: the student's score on the final exam

Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \qquad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \quad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\widehat{\beta}_1 = 11.07$ and $\widehat{\beta}_2 = 2.81$?

We expect that, on average, scores in some-flipped classes are 11.07 points higher than for no-flipped classes.

We expect that, on average, scores in fully-flipped classes are 2.81 points higher than for no-flipped classes.

Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2) \qquad u_i \overset{iid}{\sim} N(0, \sigma_u^2)$$

- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \quad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\widehat{\sigma}^2_u = 21.37$?

Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \qquad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \quad \widehat{\sigma}^2_u = 21.37$

> How do I interpret $\quad \widehat{\sigma}^2_u = 21.37$?

The average score varies from professor to professor by about 4.62 ($= \sqrt{21.37}$) points

Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2) \qquad u_i \overset{iid}{\sim} N(0, \sigma_u^2)$$

- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}_\varepsilon^2 = 4.25, \quad \widehat{\sigma}_u^2 = 21.37$

> How do I interpret $\widehat{\sigma}_\varepsilon^2 = 4.25$?

Interpretation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

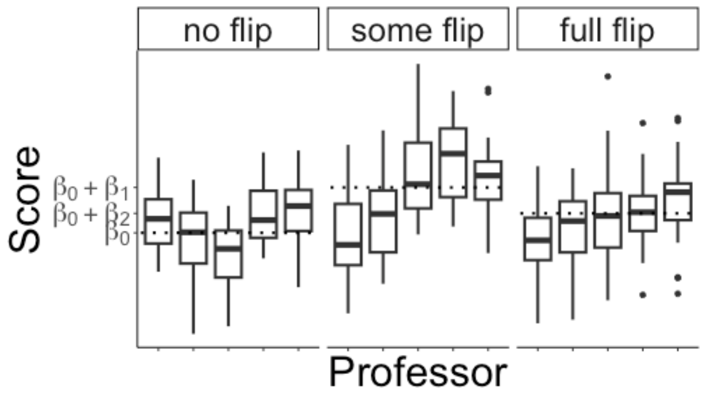
$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2) \qquad u_i \overset{iid}{\sim} N(0, \sigma_u^2)$$

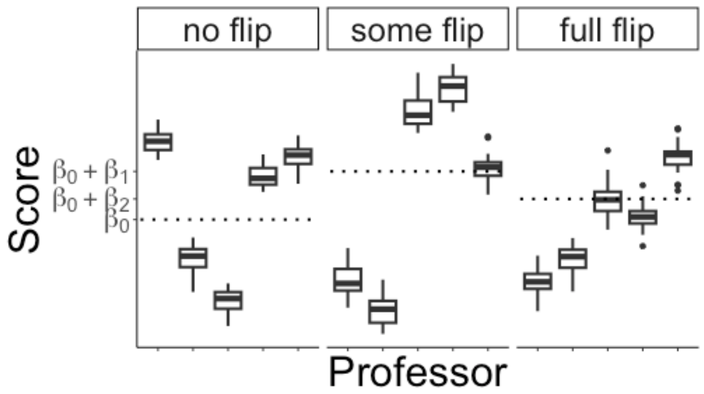
- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}_\varepsilon^2 = 4.25, \quad \widehat{\sigma}_u^2 = 21.37$

> How do I interpret $\widehat{\sigma}_\varepsilon^2 = 4.25$?

Within a class, students' scores vary by about 2.06 ($= \sqrt{4.25}$) points

Intra-class correlation





σ_ε^2 is large relative to σ_u^2

σ_ε^2 is small relative to σ_u^2

- Observations within a group are *more correlated* when σ_ε^2 is small relative to σ_u^2
- **Intra-class correlation:**

$$\rho_{group} = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2} = \frac{\text{between group variance}}{\text{total variance}}$$

Intra-class correlation

$$Score_{ij} = \beta_0 + \beta_1 \text{SomeFlipped}_i + \beta_2 \text{FullyFlipped}_i + u_i + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon) \qquad u_i \overset{iid}{\sim} N(0, \sigma^2_u)$$

- $\widehat{\beta}_0 = 77.66, \quad \widehat{\beta}_1 = 11.07, \quad \widehat{\beta}_2 = 2.81$
- $\widehat{\sigma}^2_\varepsilon = 4.25, \quad \widehat{\sigma}^2_u = 21.37$

$$\hat{\rho}_{group} = \frac{21.37}{21.37 + 4.25} = 0.83$$

So 83% of the variation in student's scores can be explained by differences in average scores from class to class (after accounting for teaching style). That's huge!

Class activity

Work on the class activity (handout).

Class activity

Interpret the estimate fixed effect coefficients $\widehat{\beta}_0$ and $\widehat{\beta}_1$.

Class activity

> Interpret the estimate fixed effect coefficients $\widehat{\beta}_0$ and $\widehat{\beta}_1$.

On average (across neighborhoods), we expect that the price of rental with 0 overall satisfaction is $27.28.

For a fixed neighborhood, an increase of 1 point in overall satisfaction is associated with an increase of $14.81 in rental price.

Class activity

> Calculate and interpret the estimated intra-class correlation.

Class activity

> Calculate and interpret the estimated intra-class correlation.

$$\hat{\rho}_{group} = \frac{1048}{1048 + 6762} = 0.134$$

About 13% of the variability in price can be explained by differences in the average price between neighborhoods (after accounting for overall satisfaction).

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$u_i \stackrel{iid}{\sim} N(0, \sigma^2_u) \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2_\varepsilon)$

where $Price_{ij}$ is the price of rental j in neighborhood i

> What assumptions are we making in this mixed effects model?

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

- **Shape:**
  - the overall relationship between satisfaction and price is linear
  - The slope is the *same* for each neighborhood

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

- **Independence:**
  - random effects are independent
  - observations within neighborhoods are independent after accounting for the random effect (i.e., the random effect captures the correlation within neighborhoods)
  - observations from different neighborhoods are independent

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$

- **Normality:** Both $u_i \sim N(0, \sigma_u^2)$ and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$
- **Constant variance:**
  - ε_{ij} has the same variance σ_ε^2 regardless of satisfaction or neighborhood

Checking assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$

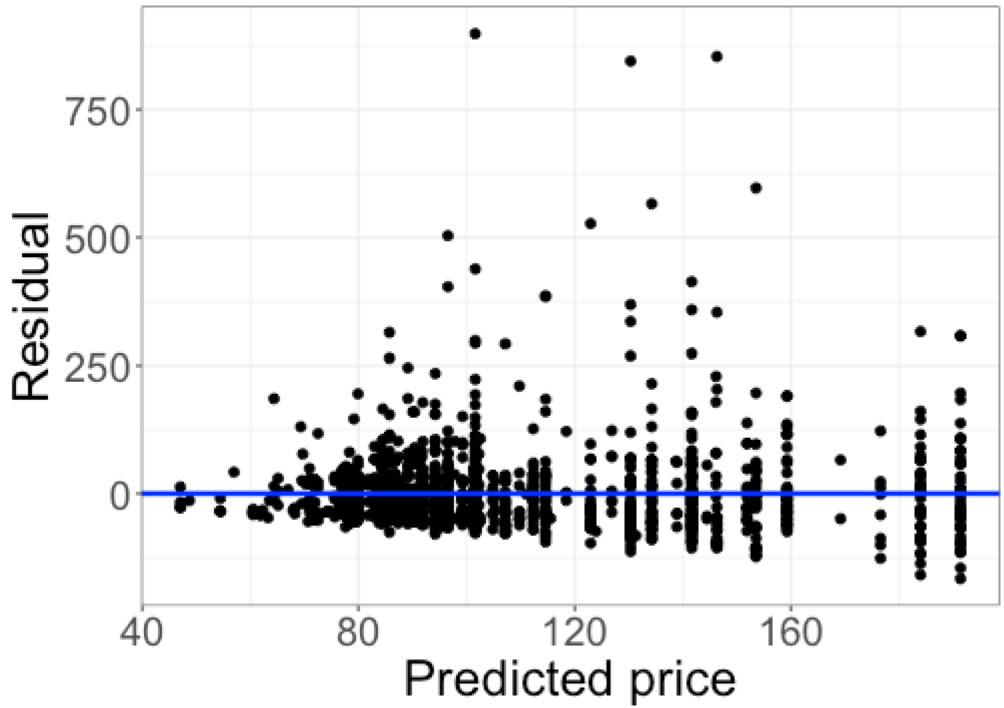
- **Shape assumption:**
  - the overall relationship between satisfaction and price is linear
  - The slope is the *same* for each neighborhood
- **Constant variance assumption:**
  - ε_{ij} has the same variance σ_ε^2 regardless of satisfaction or neighborhood

> How do you think we could check the shape and constant variance assumptions?

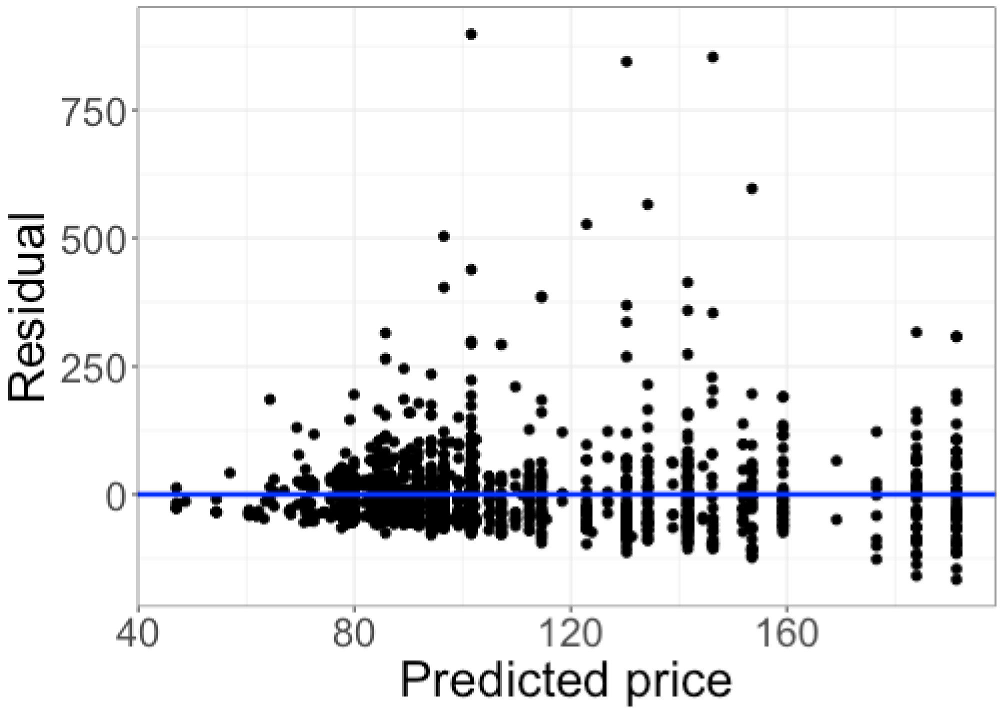
Residual plots

Residuals: $Price_{ij} - \widehat{Price}_{ij}$, where

$$\widehat{Price}_{ij} = \widehat{\beta}_0 + \widehat{\beta}_1 Satisfaction_{ij} + \widehat{u}_i$$

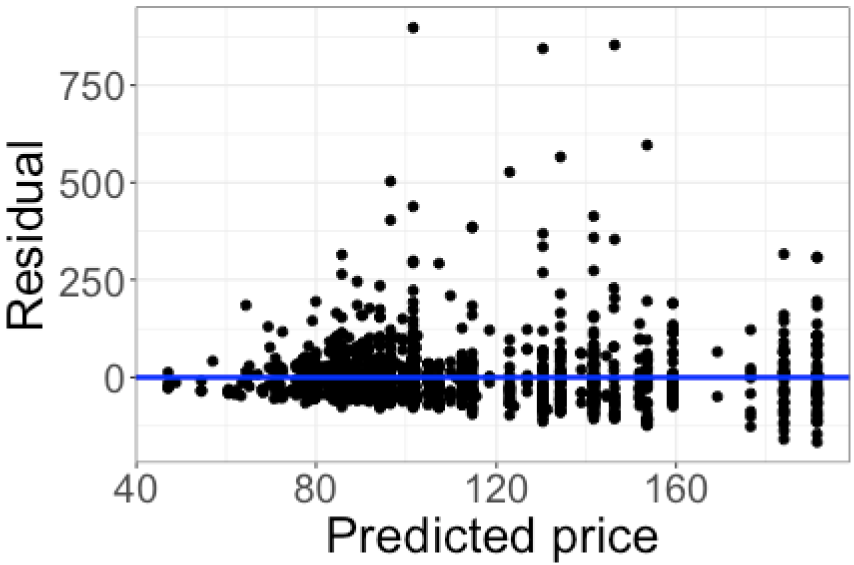


Residual plots



Do the shape and constant variance assumptions look reasonable?

Residual plots



- Linearity seems reasonable
- We can't assess (yet) whether the slope is actually the same for all neighborhoods
- Constant variance may be slightly violated (but probably not enough to matter)

Calculating predicted values

Residuals: $Price_{ij} - \widehat{Price}_{ij}$, where

$$\widehat{Price}_{ij} = \widehat{\beta}_0 + \widehat{\beta}_1 Satisfaction_{ij} + \widehat{u}_i$$

> Where do the $\widehat{u}_i$ come from?

Estimated random effects

R calculates an estimated random effect for each group (i.e., neighborhood):

```
m1 <- lmer(price ~ overall_satisfaction +
             (1 | neighborhood),
           data = bnb)
coef(m1)
```

```
## $neighborhood
##                      (Intercept) overall_satisfaction
## Albany Park            16.367331             14.80912
## Archer Heights          9.863461             14.80912
## Avondale               40.533851             14.80912
## Beverly                21.046464             14.80912
## Bridgeport             13.304517             14.80912
## Brighton Park          28.548742             14.80912
## Burnside               12.741349             14.80912
## Calumet Heights        11.465091             14.80912
```

Estimated random effects

```
coef(m1)
```

```
                 (Intercept) overall_satisfaction
Albany Park        16.367331             14.80912
Archer Heights      9.863461             14.80912
Avondale           40.533851             14.80912
Beverly            21.046464             14.80912
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```

What is the same for every neighborhood?

Estimated random effects

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coef(m1)
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                  (Intercept) overall_satisfaction
Albany Park         16.367331             14.80912
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Beverly             21.046464             14.80912
Bridgeport          13.304517             14.80912
```

> What is the same for every neighborhood?

The slope ($\widehat{\beta}_1$)

Estimated random effects

```
coef(m1)
```

```
                   (Intercept) overall_satisfaction
Albany Park          16.367331             14.80912
Archer Heights        9.863461             14.80912
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> What is *different* for each neighborhood?

Estimated random effects

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coef(m1)
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Albany Park        16.367331             14.80912
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Beverly            21.046464             14.80912
Bridgeport         13.304517             14.80912
```

> What is *different* for each neighborhood?

The intercept ($\widehat{\beta}_0 + \widehat{u}_i$)

Estimated random effects

```
                   (Intercept) overall_satisfaction
Albany Park          16.367331             14.80912
Archer Heights        9.863461             14.80912
Avondale             40.533851             14.80912
Beverly              21.046464             14.80912
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```

> How do I get the random effect estimates $\widehat{u}_i$?

Estimated random effects

```
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Albany Park         16.367331             14.80912
Archer Heights       9.863461             14.80912
Avondale            40.533851             14.80912
Beverly             21.046464             14.80912
Bridgeport          13.304517             14.80912
```

> How do I get the random effect estimates $\widehat{u}_i$?

The intercept for each neighborhood is $\widehat{\beta}_0 + \widehat{u}_i$

So subtract the estimate $\widehat{\beta}_0$ from the intercept for each neighborhood

Checking assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

- **Normality assumption:** Both $u_i \sim N(0, \sigma_u^2)$ and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$

> How do you think we could check the normality assumption?

QQ plots

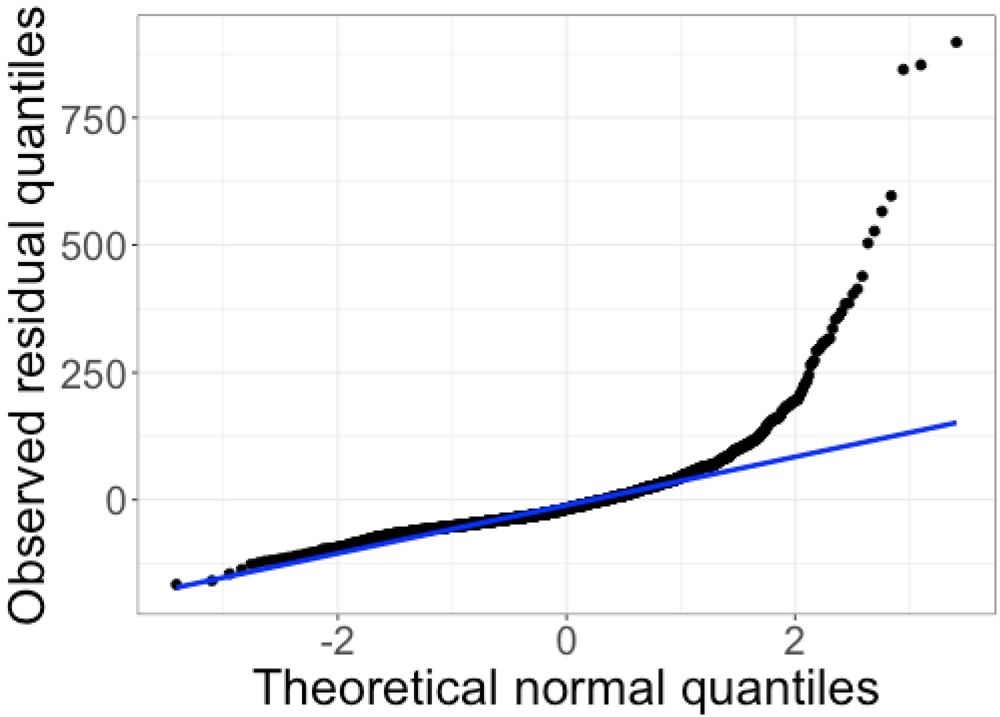
Assumption: $u_i \sim N(0, \sigma_u^2)$

- Check whether the random effect estimates $\widehat{u}_i$ appear normal with a QQ plot

Assumption: $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$

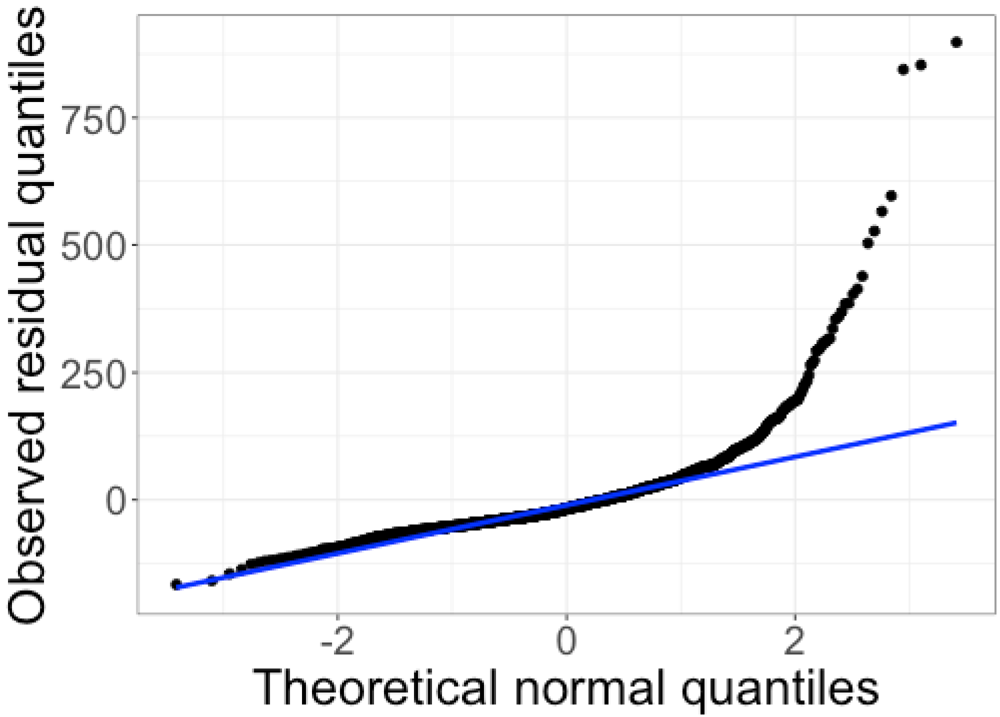
- Check whether the residuals appear normal with a QQ plot

QQ plot for the residuals



Do the residuals appear normal?

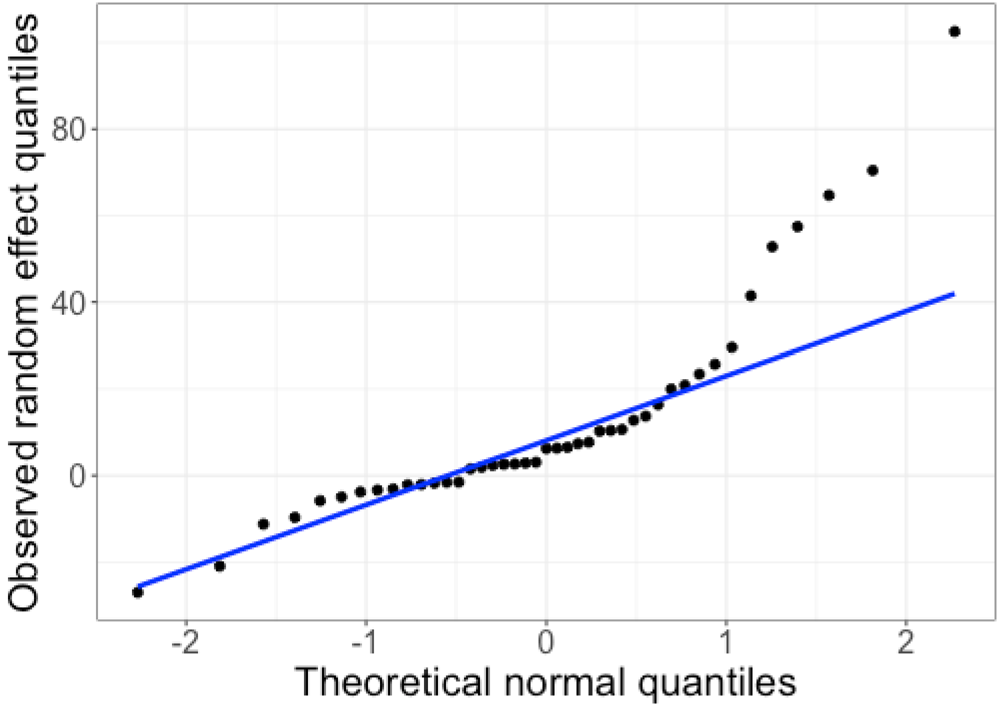
QQ plot for the residuals



> Do the residuals appear normal?

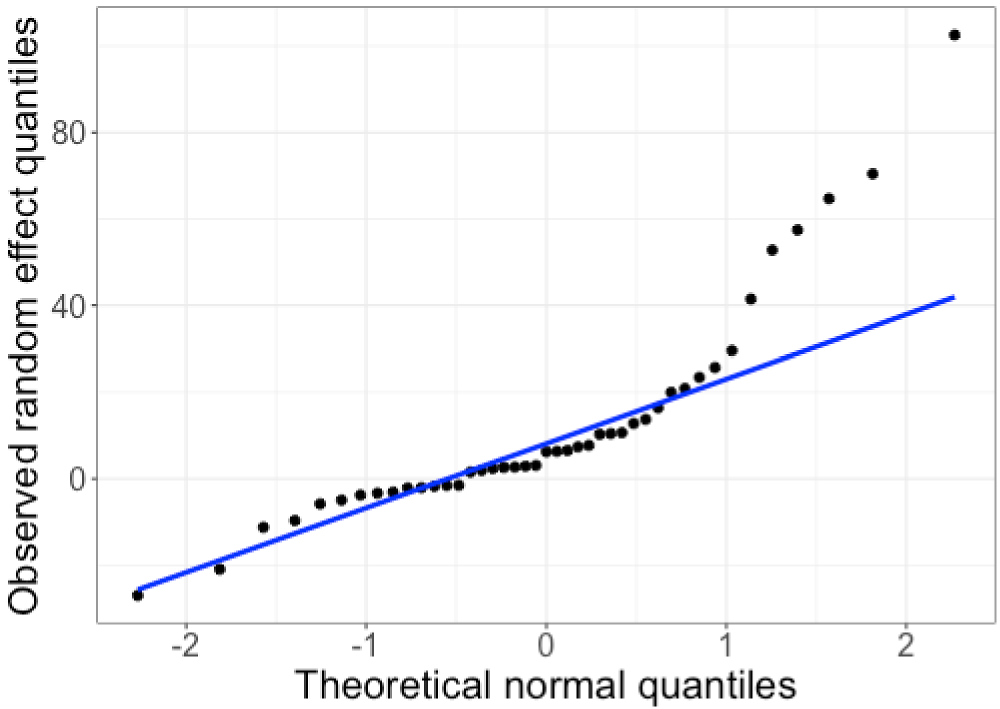
No -- though the sample size is large, so it might not matter

QQ plot for the random effects



Do the random effects appear normal?

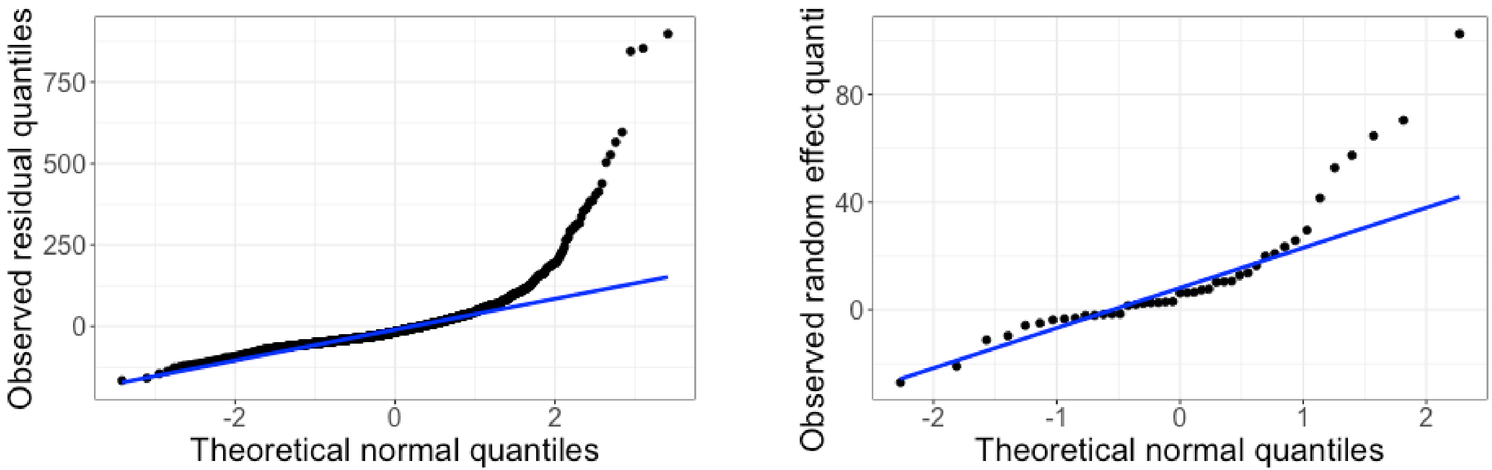
QQ plot for the random effects



> Do the random effects appear normal?

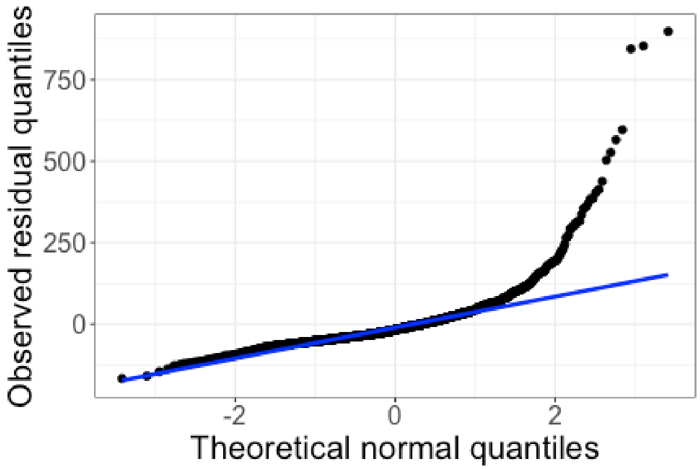
No -- and this could be more problematic

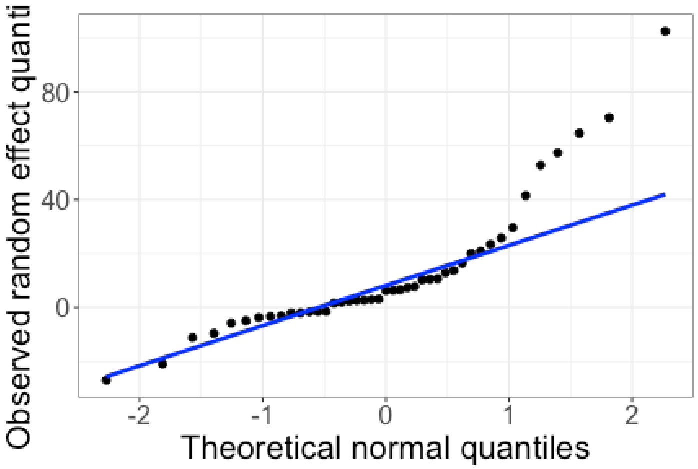
Addressing assumption violations



How could we address violations of the normality assumptions?

Addressing assumption violations





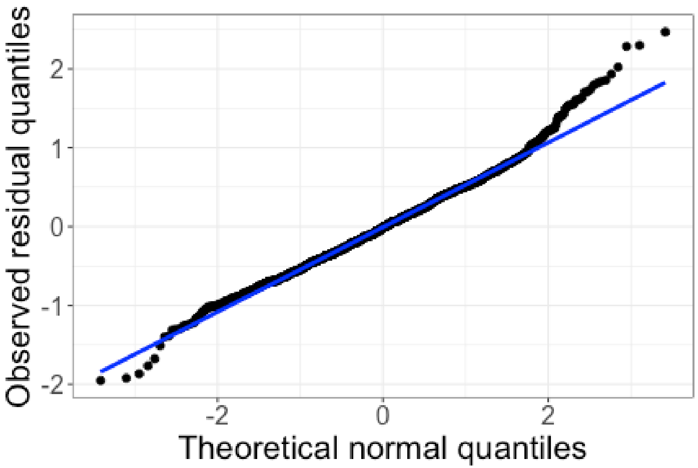
> How could we address violations of the normality assumptions?

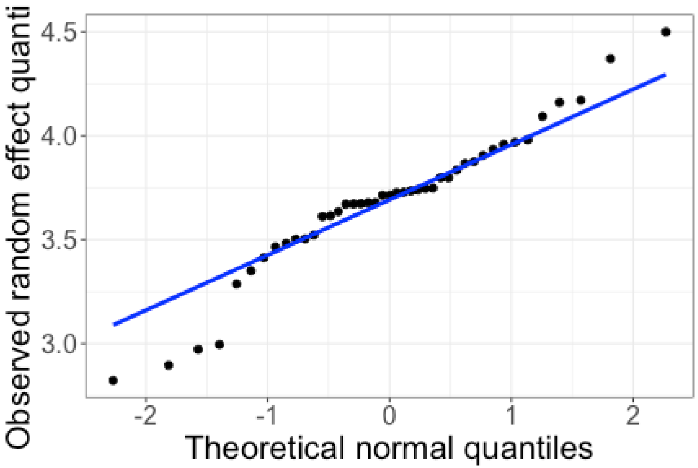
Transformations!

Transformations

$$\log(Price_{ij}) = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$





- Transformations can help with both the residuals and random effects